Project Design Phase-II | Solution Requirements (Functional & Non-functional)

Date: 27-10-2026 | **Team ID:** SWUID2026403391 | **Project Name:** Agri-Tech

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements – Agri-Tech Smart Farming System

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	Registration, Login, Password Reset JWT-based secure sessions with farmer and admin roles
FR-2	Farm Profile Setup & Management	Create and update farm profile (location, size, soil type) View and manage farm details
FR-3	Crop Monitoring & Soil Health Analysis	Add, update and track crop records and activity logs Enter soil data and view soil health reports
FR-4	Weather Forecasting & Pest Alerts	View real-time weather forecast for the farm location Receive pest alert notifications
FR-5	Resource Purchase & Order Tracking	Browse and purchase seeds, fertilizers and equipment Track order status
FR-6	Expert Consultation & Community Forums	Request advice from agricultural experts Post and reply in community forums
FR-7	Admin Dashboard	Manage users, content and resources Approve resource orders and moderate content

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The platform should offer a simple, multilingual interface that farmers of all literacy levels can use easily.
NFR-2	Security	Passwords are encrypted with bcrypt. JWT authentication and role-based access protect sensitive operations.
NFR-3	Reliability	Crop records, orders and alerts should be stored and delivered reliably under all conditions.
NFR-4	Performance	Pages should load within 3 seconds on rural networks. Weather and alert data should refresh in near real time.
NFR-5	Availability	The system should maintain 99.9% uptime across all services.
NFR-6	Scalability	Should support growing numbers of farmers and large datasets through optimized queries and modular services.